

AI CAD Copilots: Comparative Evaluation Guide

Comprehensive Technical Breakdown & Selection Matrix: CADABRA vs. Adam AI vs. AIDraft

RECOMMENDATION Executive Decision & Platform Selection

There is no single "best" copilot overall. The optimal selection depends strictly on your CAD platform ecosystem and primary engineering workflow bottleneck:

1. CADABRA — Best All-Around for SolidWorks & PDM Workflows

Native desktop add-in designed specifically for **SOLIDWORKS**. Handles end-to-end parametric modeling, assembly structure modification, PDM/PLM toolbox queries, and automated drafting. Ideal for general mechanical design engineers using Dassault systems.

2. Adam AI — Best for Cloud CAD & Parametric Tree Optimization

Optimized for cloud and web CAD ecosystems like **Autodesk Fusion** and **Onshape**. Excels at generating clean, parametric feature trees from natural language, consolidating redundant operations, and managing parameter variables.

3. AIDraft — Best for 2D Technical Drawings & Enterprise GD&T

Specialized post-modeling engine that converts completed 3D models into audit-ready 2D manufacturing drawings across **Siemens NX**, **Creo**, **SolidWorks**, and **Inventor**. Offers 100% on-premises deployment for defense and aerospace compliance.

EVALUATION Side-by-Side Architectural Feature Matrix

Feature / Metric	CADABRA	Adam AI	AIDraft
Primary Focus	End-to-End SolidWorks Assistant	Cloud Parametric Tree Generator	2D Drawing & GD&T Automation
CAD Compatibility	SOLIDWORKS (2020–2026)	Autodesk Fusion, Onshape	Siemens NX, Creo, SOLIDWORKS, Inventor
Deployment	Windows C# .NET COM Add-In	Cloud / Native Web API Plugin	On-Premises / Air-Gapped or Plugin
Output Format	Parametric <code>.SLDPRT</code> , <code>.SLDASM</code>	Cloud B-Rep <code>.F3D</code> , Onshape Tree	2D Vector Prints (<code>.DWG</code> , <code>.DXF</code> , PDF)
PLM / PDM Hook	Direct PDM Professional / Enterprise API	Cloud Workspace Metadata API	Title Block & Revision Table Connectors
Execution Latency	1.5s – 5.0s per modeling operation	2.0s – 6.0s per feature tree build	10s – 25s for full 2D sheet layout

SPECIFICATION Underlying AI Copilot Infrastructure

- **Multi-Agent Orchestration:** Requests pass through an Orchestrator/Router that delegates tasks to specialized sub-agents (Code Synthesis, Geometry Topology, PDM Standards, and Rebuild Error Healing).
- **Code Synthesis over Mesh:** Systems compile programmatic C# COM, C++, or Python API calls rather than returning uneditable mesh objects (e.g. `.STL` / `.OBJ`).
- **Context Window Allocation:** Employs 128k to 200k input token context limits to load entire assembly feature trees, company drawing standards, and BOM JSON models simultaneously.